

FAT TAILS, INSTITUTIONAL FLOW, AND THE LIMITS OF VOLATILITY SIGNALS: DENSITY FORECASTING EVIDENCE FROM CUSTODIAN-LEVEL FOREIGN INSTITUTIONAL TRADING IN INDIA

Abstract

Purpose: This study examined whether the composition of foreign institutional investor (FII) trading carried information about the risk of Indian equities that standard volatility models did not already contain, and whether any such information survived conversion into out-of-sample density forecasts.

Design/Methodology/Approach: Fourteen years of masked custodian-level FII transaction records (January 2011 to March 2025, approximately 24 million trades) were merged with a corporate-action-adjusted price panel. A gap-aware hidden Markov model classified daily flow composition, and an empirical, past-only conditional density engine was estimated walk-forward under a strict information embargo. Forecasts were scored using the continuous ranked probability score, Kupiec and Christoffersen backtests, probability integral transform uniformity, and Diebold–Mariano tests. Four hypotheses were pre-registered before estimation, with pass criteria fixed in advance.

Findings: Flow composition added nothing to stock-day density forecasts. The same engine nevertheless outperformed the Gaussian benchmark decisively: per-stock Kupiec pass rates were 90.4% against 60.4%, because standardized losses beyond four sigma occurred 85 times more often than a Gaussian implied. Two flow effects survived. Institutional share of turnover predicted next-day volatility, and aggregate FII inflow improved market-level risk forecasts, but the effect vanished against a competitive asymmetric GARCH benchmark.

Practical Implications: Gaussian value-at-risk systems materially understate Indian equity tail risk. Regression significance is a weak guide to forecasting value.

Originality/Value: Composition-based, pre-registered density evaluation of institutional flow in an emerging market.

Keywords: density forecasting, foreign institutional investors, value-at-risk, tail risk, pre-registration, emerging markets

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INTRODUCTION

Foreign institutional investors have been the most closely watched marginal participant in Indian equities for three decades. Regulators publish their aggregate daily net purchases, financial media treat those figures as a sentiment barometer, and a substantial literature reports that FII flows co-move with returns and volatility. Almost all of that literature works with a single aggregate rupee number per day. The underlying regulatory records, however, are far richer: every trade is reported at the custodian level, so the *composition* of institutional trading, that is, how many distinct institutions were active, how concentrated their participation was, and whether they were buying or selling, is observable in principle.

This study asked whether that composition carried risk information beyond what a conventional volatility model already knew. The question matters because it separates two claims that are routinely conflated. The first is that flow correlates with subsequent volatility, which is a

statement about a regression coefficient. The second is that flow improves a risk forecast, which is a statement about an out-of-sample loss function and is what a risk manager actually needs. The distance between the two claims is the central concern of this paper.

The study was designed so that a negative answer would be as informative as a positive one. Four hypotheses were pre-registered with pass criteria fixed in advance, and every attempt is reported here, including three that failed. The evaluation used proper scoring rules rather than significance tests alone, because a proper scoring rule is the only instrument that answers the question a practitioner asks.

Three results emerged. Flow composition added nothing to stock-day density forecasts, a finding established with power to detect very small effects. The forecasting machinery built to test that hypothesis nevertheless outperformed the standard Gaussian benchmark by a wide margin, for a reason unrelated to institutional flow: Indian equity returns, even after standardization by a conditional volatility estimate, are far more extreme than a normal distribution admits. Finally, two genuine flow effects were isolated at different levels of aggregation, one cross-sectional and one at the market level, and the boundary at which each stopped being useful was measured precisely.

REVIEW OF LITERATURE

Institutional Flow and Emerging Market Prices

The foundational work on foreign participation in emerging markets established that liberalization affected the cost of capital and volatility structure of local markets (Bekaert & Harvey, 2000). Subsequent research documented that international investor flows exhibit strong persistence and that portfolio flows into emerging markets responded to, and partially forecast, local returns (Froot et al., 2001). For India specifically, the relationship between foreign flows and volatility has been examined repeatedly. Foreign institutional investment has been linked to stock market volatility using conditional heteroskedasticity models (Dadhich et al., 2015), asym-

metric volatility responses to portfolio flows have been documented (Varughese & Mathew, 2017), and the association between institutional flows and implied volatility has been examined in the derivatives segment (Sharma et al., 2023). These studies share two features: they use aggregate flow magnitudes, and they evaluate in-sample coefficients rather than out-of-sample forecasts.

A parallel literature examined institutional trading at the transaction level, finding that institutional herding and correlated trading affect prices, and that the price impact of institutional trades separates into transitory and permanent components (Sias, 2004). The microstructure tradition provides the identification logic: the probability of informed trading can be estimated from order-flow counts alone (Easley et al., 1996), which offers a route to distinguishing information-driven from liquidity-driven institutional activity without observing prices.

Volatility Modelling and Density Evaluation

The benchmark against which any volatility claim must be judged is well established. Generalized autoregressive conditional heteroskedasticity (Bollerslev, 1986) and its asymmetric extension, which allows negative returns to raise conditional variance more than positive ones (Glosten et al., 1993), remain difficult to beat. Comparative evidence suggests that a simple GARCH(1,1) is rarely outperformed by more elaborate specifications in out-of-sample forecasting (Hansen & Lunde, 2005), which sets a high bar for any claimed improvement.

Evaluation methodology is equally settled. Interval forecasts are assessed through unconditional coverage (Kupiec, 1995) and conditional coverage tests that detect clustering of violations (Christoffersen, 1998). Full predictive densities require strictly proper scoring rules, of which the continuous ranked probability score is the standard choice (Gneiting & Raftery, 2007). Comparisons between competing forecasts rest on tests of equal predictive accuracy (Diebold & Mariano, 1995), implemented with heteroskedasticity-and-autocorrelation-consistent standard errors (Newey & West, 1987).

Tail Risk

A recurring finding is that conditional normality fails in the tails even after volatility standardization. Extreme value methods applied to heteroskedastic financial series produce materially different tail estimates from Gaussian ones (McNeil & Frey, 2000), and the practical consequence is that Gaussian value-at-risk systems understate capital requirements in exactly the region where losses concentrate.

The Research Gap

Three gaps motivated this study. First, the Indian FII literature works almost exclusively with aggregate flow magnitudes; the composition of that flow has not been evaluated as a risk variable. Second, that literature reports regression coefficients rather than out-of-sample density performance, so it is unknown whether reported significance would survive conversion into a forecast. Third, pre-registration is essentially absent from this literature, which makes published effect sizes difficult to interpret given the number of specifications available to a researcher.

DATA AND METHODOLOGY

Data

The primary source was a masked custodian-level record of foreign institutional transactions in Indian equities covering January 2011 through March 2025, comprising approximately 24 million trades. Each record carries a masked entity identifier, security identifier, trade date, direction, quantity, and value. Only equity buy and sell trades with a positive rate were retained. These were merged with a daily price panel constructed from exchange bhavcopies and adjusted for corporate actions, together with the India VIX, the Nifty 50 index, and the rupee–dollar exchange rate.

The sample was split at a frozen boundary. The training era ended 30 April 2021 and the test era began 1 July 2021, with the intervening two months excluded because the source feed contains no records for them. This split was fixed before any modelling and was never revisited. The

training era contains 508,563 stock-days across 618 securities; the test era contains 294,243 stock-days across 851 securities. The test era is therefore a different universe rather than a repeated sample, which strengthens any out-of-sample claim.

Three further months (June, September, and November 2023) carry a completely null trade-direction field in the source archive, affecting 3.0% of trades. These were excluded rather than imputed, since imputing direction would fabricate the dependent construct of the study. All exclusions fall inside the test era and are reported rather than silently dropped.

Flow Composition and the State Model

Daily flow features were computed within each stock-day and rank-normalized across the cross-section, which by construction removes level information and isolates composition. A three-state Gaussian hidden Markov model in the regime-switching tradition (Hamilton, 1989) was estimated on these features. Because the FII feed contains gaps of varying length, the transition matrix was raised to the power of the number of elapsed trading days between observations, so that state belief decays correctly across holes rather than advancing one step per observation. Model parameters were re-estimated walk-forward, producing a sequence of parameter vintages, each of which used only data available at its own as-of date.

The Density Forecasting Engine

Let $r_{i,t+h}$ denote the h -day forward return of stock i and $\sigma_{i,t}$ an exponentially weighted volatility estimate computed from past returns only. The standardized outcome is

$$z_{i,t+h} = \frac{r_{i,t+h}}{\sigma_{i,t} \sqrt{h}}. \quad (1)$$

The engine estimates the conditional density of z non-parametrically from matured past outcomes, weighted by the posterior probability that stock i occupied each flow-composition state on day t :

$$\hat{f}_{i,t}(z) = \sum_{a=1}^A \pi_{i,t}(a) \hat{f}_a(z), \quad (2)$$

where $\pi_{i,t}(a)$ is the filtered posterior over archetypes and $\hat{f}_a$ is the empirical density of standardized outcomes historically associated with archetype a . Every outcome entering $\hat{f}_a$ satisfies $s + h \leq \text{asof}$, so no forecast can use an outcome that had not matured at the time the forecast was made. The benchmark replaces $\hat{f}_{i,t}$ with the standard normal, giving the conventional exponentially weighted moving average value-at-risk engine used throughout the industry.

The critical control is a variant in which the archetype weights are replaced by their unconditional climatological values, holding all other machinery fixed. Comparing the conditional engine against this control isolates the contribution of flow information from the contribution of distributional shape.

Market-Level Engine

At the market level the density was specified parametrically. Next-day Nifty returns were modelled as $r_{t+1} = s \cdot m_t \cdot \sigma_t \cdot T_v$, where T_v is a standardized Student- t variate, σ_t is a conditional volatility estimate, and the tilt is

$$m_t = \exp(\theta' x_t), \quad m_t \in [0.5, 2.0], \quad (3)$$

with x_t a vector of conditioning variables. All parameters were estimated by maximum likelihood on the expanding past, first fitted after 750 scored days, refitted every 21 trading days, and applied out of sample to the following block. Aggregate flow entered at a two-day lag throughout, reflecting the publication delay of the underlying disclosure, so every specification is deployable in real time.

Evaluation Protocol and Pre-Registration

Forecasts were evaluated using the continuous ranked probability score, pinball loss at the 5% and 1% quantiles, Kupiec unconditional coverage tests, Christoffersen independence tests, and probability integral transform uniformity assessed against a date-block bootstrap null constructed with a stationary resampling scheme (Politis & Romano, 1994) rather than a textbook chi-square, which is invalid under cross-sectional dependence. Differences in predictive accuracy were tested by Diebold–Mariano statistics with Newey–West standard errors.

Four hypotheses were pre-registered in dated documents written before estimation. Each specified the exact variable construction, the control set, the model comparison, and a numerical pass bar requiring both full-sample significance and same-signed consistency across the two eras. Every pre-registered attempt is reported in the results, including those that failed.

RESULTS AND DISCUSSION

Flow Composition Adds Nothing to Stock-Day Densities

The primary hypothesis was rejected. The archetype-conditional mixture was statistically indistinguishable from the identical machinery with conditioning removed, at every horizon and in every stratum examined, including strata selected for high flow uncertainty where any conditioning effect should be largest. The comparison was powered to detect differences above roughly 0.03% of the score. Composition, as measured here, carries no incremental information about the shape of the conditional return distribution once volatility is controlled.

This is a clean negative result rather than an inconclusive one, and it constrains the interpretation of the aggregate-flow literature: whatever information FII activity carries, it does not reside in the cross-sectional composition of daily trading at the individual stock level.

The Gaussian Benchmark Fails in the Tail

The engine nevertheless outperformed the benchmark decisively, as Table 1 shows, and the

reason has nothing to do with flow.

Table 1. Density Forecast Performance, Empirical Engine Versus Gaussian Benchmark, One-Day Horizon

Evaluation criterion	Empirical engine	Gaussian benchmark
Breach rate at 1% level (target 1.00%)	1.04%	1.62%
Kupiec test p -value at 1% level	0.673	0.000
Probability integral transform statistic	0.7	303.3
Per-stock Kupiec pass rate	90.4%	60.4%
Expected shortfall coverage (target 1.000)	1.027	1.246
Mean continuous ranked probability score	0.55785	0.56095

Note. Walk-forward out-of-sample results on 578,481 stock-days covering 562 securities, February 2016 to March 2025. Both engines share an identical volatility estimate and identical evaluation dates; they differ only in the assumed shape of the standardized distribution. Source: Authors' calculations.

The mechanism appears in Table 2. Standardized losses, that is, losses already scaled by a conditional volatility estimate, occur at the extremes far more often than a normal distribution permits.

Table 2. Observed and Gaussian-Implied Frequencies of Extreme Standardized Losses

Threshold	Observed count	Gaussian expectation	Ratio
Below -4 standard deviations	1,562	18.3	85
Below -5 standard deviations	742	0.17	4,475
Below -6 standard deviations	398	0.00	Approx. 700,000
Below -8 standard deviations	133	0.00	Not defined

Note. Counts computed on the same 578,481 stock-days as Table 1. Source: Authors' calculations.

A Gaussian engine is not marginally conservative in this market; it is wrong by orders of magnitude in precisely the region where capital is destroyed. The empirical engine estimated the ratio of expected shortfall to value-at-risk at 1.39 to 1.41, against the 1.146 that a Gaussian imposes by construction regardless of market conditions. The engine did fail at the twenty-day horizon, where the breach rate reached 12.4%, and it is not presented as a long-horizon tool.

Institutional Share of Turnover Predicts Volatility

A distinct hypothesis concerned the level rather than the composition of institutional participation: whether the share of a stock's turnover attributable to institutional trading predicts its

next-day volatility. Table 3 reports the result, with the squared standardized outcome as the dependent variable, so the test asks whether the share explains volatility that the engine's own estimate has already failed to capture.

Table 3. Institutional Share of Turnover and Next-Day Volatility

Specification	Full sample	Training era	Test era
Date fixed effects	−0.0283 (−5.25)	−0.0370 (−4.43)	−0.0198 (−2.85)
Stock and date fixed effects	−0.0300 (−5.53)	−0.0365 (−4.42)	−0.0232 (−3.25)
Adding own lagged volatility	−0.0255 (−4.59)	−0.0332 (−3.87)	−0.0180 (−2.50)
Share observable only at two-day lag	−0.0015 (−0.32)	−0.0020 (−0.30)	−0.0008 (−0.13)

Note. Dependent variable is the squared standardized next-day return. Sample comprises 577,245 stock-days, 972 securities, and 2,081 trading dates. Date-clustered t -statistics in parentheses. Logarithm of turnover is included as a control in every specification and enters with a t -statistic above 24 throughout. Source: Authors' calculations.

The effect is negative, statistically strong, and present in both eras under the demanding two-way fixed-effects specification. Stock-days with a lower institutional share of turnover are subsequently more volatile. Sorting on the within-day quintile of institutional share produced a monotone gradient in implied volatility from 1.0911 in the lowest quintile to 1.0357 in the highest, a spread of 5.3%. The result is robust to the choice of winsorization threshold, is stronger when absolute rather than squared standardized returns are used, and is stronger in high-volatility regimes.

The final row of Table 3 is the finding's own limit. When the share is lagged to the point at which it would genuinely be observable to a market participant, the coefficient collapses to zero. The effect is therefore economic rather than tradeable, and it is reported as such.

Aggregate Flow Improves Market-Level Risk Forecasts

The one deployable positive result concerns aggregate flow at the index level. A pre-registered regression established that aggregate net FII inflow observed at a two-day lag predicted lower absolute next-day Nifty returns beyond the information already in the India VIX, with a t -statistic

of -4.78 on the full sample and -2.49 and -2.51 in the two eras separately, clearing the pre-registered bar.

A natural objection is that this reflects the leverage effect, since inflows follow rising markets and rising markets precede lower volatility. The objection was tested directly by adding signed-return controls. The coefficient retained a t -statistic of -4.23 on the full sample and remained significant in both eras, and past signed returns explained only 2.8% of the variation in the flow variable. The signal is incremental information, not a repackaged leverage effect.

Converting the regression into a forecasting engine produced the results in Table 4. All three pre-registered conditions were satisfied.

Table 4. Pre-Registered Gate for the Market-Level Flow Engine

Condition fixed in advance	Requirement	Outcome
Predictive accuracy against a volatility-index benchmark	Diebold–Mariano statistic at or below -2.0	-2.35 (satisfied)
Consistency across eras	Lower mean score in both training and test eras	-0.0019 and -0.0004 (satisfied)
Interval calibration preserved	Kupiec p -value above 0.01 at both levels	0.225 and 0.161 (satisfied)

Note. Based on 2,739 scored trading days. All parameters were estimated walk-forward on past data only. Source: Authors' calculations.

The economic magnitude was modest but real. At matched one-percent coverage, the flow-augmented engine required approximately 2% less capital, and the average loss in excess of the value-at-risk threshold fell from 0.770% to 0.613%, an improvement of roughly one-fifth. The gain was monotone in the signal, being negligible on quiet days and largest on days of strong inflow, and it concentrated in the one-percent tail rather than in the centre of the distribution.

The Boundary: Where the Signal Stops Converting

Three further pre-registered attempts extended the engine, and all three failed. Table 5 summarizes them, because a boundary measured from both sides is more informative than a single success.

Three lessons follow. First, the dollar index and the policy-rate differential added nothing once

Table 5. Pre-Registered Engine Extensions and Their Outcomes

Extension	Question tested	Outcome
Asymmetric GARCH base	Does flow survive a competitive volatility benchmark?	Not satisfied (−0.95)
Global conditioning variables	Do exchange rate, United States equity, dollar index, and policy rate differentials help?	Not satisfied (−0.67)
Weekly horizon	Does a five-day horizon recover the signal?	Screen satisfied, engine not satisfied (+1.58)

Note. Entries report Diebold–Mariano statistics against the relevant benchmark; negative values favour the augmented model, and the pre-registered threshold was −2.0 in every case. Source: Authors’ calculations.

the rupee exchange rate was included, indicating that the India-relevant component of global dollar conditions is already impounded in the currency. Second, the weekly spillover from United States equity returns to Indian volatility was the most era-stable coefficient estimated anywhere in this study, with t -statistics of −3.50, −2.72, and −4.39 on the full sample and the two eras, and yet it produced no density-forecast improvement whatsoever, because an asymmetric GARCH benchmark already absorbs the shock through the index’s own lagged returns.

Third, and most generally, the same pattern recurred four times with four different signals. Conditioning variables with regression t -statistics between three and five failed to improve a well-specified conditional density forecast, and succeeded only against a weak benchmark. A modulation of conditional volatility of one to two percent is genuine, survives every control tested, and still lies below the detection threshold of a proper scoring rule over roughly 2,700 observations. This is the distance between significance and forecasting value, and it is large.

Robustness and Internal Audit

Two classes of check were applied. First, a look-ahead audit re-executed the entire forecasting chain against inputs truncated at each as-of date and confirmed that pre-truncation and post-truncation quantities agreed exactly, verifying that no forecast used unavailable information.

Second, an internal review of the estimation code identified two defects and both were corrected before the results reported here were finalized. One concerned models whose conditioning parameters were applied out of sample before sufficient estimation history had accumulated, which asymmetrically penalized parameter-rich specifications; a minimum-history rule was imposed symmetrically across all models. The other concerned optimizer path-dependence on a flat likelihood surface, which was resolved by tightening convergence tolerances and adding a derivative-free polish step. The market-level engine result in Table 4 was unchanged to every reported digit under both corrections, whereas two apparent near-misses widened into clear failures, and one apparent recovery of a positive result proved to be an artifact and was discarded. This experience is reported because it illustrates concretely how small claimed edges can be manufactured or destroyed by estimation choices that are rarely documented.

Finally, no strategy constructed from any of these signals was profitable after realistic transaction costs. The best test-era gross Sharpe ratio was 1.51, which became -1.53 at a fifteen basis point round-trip cost. Costs bind, and that is itself a result.

CONCLUSION

This study asked whether the composition of foreign institutional trading carried risk information in Indian equities, and answered in the negative at the level of individual stock-day return distributions, with power to detect very small effects. That negative result is credible precisely because it was pre-registered and because the same machinery produced strong positive findings elsewhere.

The most consequential finding was incidental to the original question. Standardized Indian equity returns are so far from Gaussian in the tails, with four-sigma losses occurring 85 times more frequently than a normal distribution implies, that an empirical conditional density estimated from past outcomes alone raised the proportion of securities passing a standard coverage test from 60.4% to 90.4%. Two genuine flow effects were isolated: institutional share of turnover

predicts cross-sectional volatility but not at a deployable lag, and aggregate inflow improves market-level risk forecasts by about 2% of capital, though not against an asymmetric GARCH benchmark.

The unifying contribution is a measurement of the distance between statistical significance and forecasting value. Across four independent signals, that distance was large and systematic.

RESEARCH IMPLICATIONS

For practitioners, the first implication is direct. Institutions using Gaussian value-at-risk systems on Indian equities are understating tail risk substantially, and the correction requires no new data, only the replacement of a distributional assumption with an empirical one estimated from past standardized outcomes.

For risk managers at the index level, aggregate FII flow is a usable conditioning variable that improves tail calibration at a genuinely deployable lag, provided the underlying volatility model is simple. Institutions already running asymmetric GARCH systems should not expect a gain.

For researchers, the implication is methodological. A regression coefficient significant at conventional levels is weak evidence that a variable will improve a forecast. Studies claiming predictive content for volatility should report out-of-sample density performance against a competitive benchmark, and pre-registration materially improves the interpretability of the resulting effect sizes.

For regulators, the institutional share of turnover is a screening variable constructed entirely from data already collected, associating lower institutional participation with elevated subsequent volatility.

LIMITATIONS OF THE STUDY AND SCOPE FOR FURTHER RESEARCH

Several limitations qualify these findings. The transaction records are masked at the entity level, so institution-specific characteristics such as mandate, domicile, or investment style cannot be observed, and heterogeneity across institutional types could not be tested. The records contain no intraday timestamps, so trade aggressiveness and execution timing, which the microstructure literature identifies as central to price impact, are outside the scope of this study.

Five months of flow data are unavailable, two masked at source and three carrying a null direction field, together representing approximately 3% of trades and falling entirely within the test era. These were excluded rather than imputed, which is conservative but reduces test-era power.

The forecasting results apply to end-of-day risk measurement at horizons of one and five trading days; the engine is not calibrated at twenty days and should not be deployed there. All results concern a single emerging market over a single fourteen-year window that includes one extreme volatility episode.

Four extensions appear promising. Intraday data would permit the aggressiveness and timing dimensions to be tested directly. Unmasked or partially classified entity identifiers would allow institutional heterogeneity to be examined. Replication in other emerging markets with comparable custodian-level disclosure would establish whether the tail findings generalize. Finally, the systematic gap between regression significance and density-forecast improvement documented here deserves study in its own right, since it bears on a large body of published volatility research.

AUTHOR'S CONTRIBUTION

[INSERT NAME] conceived the research question, designed the empirical strategy, constructed the data pipeline, wrote and audited all estimation code, pre-registered the hypotheses, inter-

preted the results, and prepared the manuscript.

CONFLICT OF INTEREST

The author certifies that there is no conflict of interest with any financial or non-financial interest in the subject matter discussed in this manuscript.

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